RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year - Semester II Examination – March - May 2026

COM3307 – EMBEDDED SYSTEMS

Time: Three (03) hours

- This paper consists of **Five (05)** questions in **Two (02)** pages.
- Questions 1 and 2 are mandatory. Answer any 2 questions from Questions 3 to 5.

---

1. a) Define embedded systems and explain three (03) key characteristics that differentiate it from general-purpose computing systems. (06 marks)

   b) Explain the difference between functional requirements and non-functional requirements with one (01) embedded system example. (08 marks)

   c) A company is designing a smart home temperature monitoring system.
   Analyze the system and identify:
   - Inputs
   - Outputs
   - Processing unit
   - Two constraints

   Justify your answers. (11 marks)

2. Answer questions a) – c) based on the following scenario.

   A smart street lighting system automatically turns lights on at night and adjusts brightness based on ambient light conditions.

   a) Identify suitable:
   - Sensors
   - Actuators
   - Controller (06 marks)

   b) Describe the system logic using pseudocode or flowchart-style steps. (09 marks)

c) Evaluate two (02) improvements to make the system more efficient or reliable. (10 marks)

3. a) Consider the following code snippet:

```
ldi r16, 0x05
ldi r17, 0x03
sub r16, r17
```

What is the final value in `r16`? Briefly explain which flags may be affected. (12 marks)

b) Analyze the following code snippet:

```
    ldi r18, 0x03
Loop:
        dec r18
        brne Loop
```

What does this code do? Explain your reasoning. (13 marks)

4. a) Write an AVR assembly code snippet to:
   - Configure `PORTB` as output
   - Send value `0xAA` to `PORTB` (Use your knowledge in `DDRx`, `PORTx` and `PINx`)

   (10 marks)

b) A system reads a switch from `PORTD` and turns ON an LED connected to `PORTB` when the switch is pressed. Write a simple logic (assembly or pseudocode) to achieve this. (15 marks)

5. a) Describe how an instruction is executed in a microcontroller using the fetch–decode–execute cycle. Illustrate your answer with a simple example instruction. (10 marks)

b) You are asked to model a simple microcontroller architecture.
Design a block diagram (or describe clearly) including:
   - CPU
   - Program Memory
   - Data Memory
   - Registers
   - Input/Output ports

Then explain how data flows through the system when a value is:
   - read from an input port.
   - processed by the CPU.
   - sent to an output port.

(15 marks)

--- END ---